Screened NR transport within the existing ROOT budget

Branch: `g4v11.3_cygno04_screening`, 21 October 2026. Original control: `g4v11.3_cygno04`, commit `3f53d24ce433e304dd86f31ef9c5f16cefd50ee1`. All builds/runs reported here used **Geant4 11.4.0 and ROOT 6.36.08 under WSL**. The historical branch name does not establish a tested Geant4 11.3 build.

Technical PDF, detailed comparison PDF and comparison CSV describe the current implementation. They supersede the earlier fully tracked screening results for this branch.

Required output contract

The original ROOT format is preserved, with one hit per native depositing Geant4 step. Branch names, types, units and vector alignment match the original. There is no aggregation, packing, quantization, hit thinning or new reader. The ROOT writer, sensitive detector, event action and primary generator are identical to the original branch. `numhits` still counts native depositing steps. Raw energies are never rescaled to enforce quenching.

The target is 10 decimal GB per ion per million events of the existing 0.25-1000 keV log-uniform NR source. The earlier fully tracked screening prototype required approximately 37-572 GB/million in this format. That is not the model now installed. The rejected compact-output implementation is absent from this branch.

Physics model and compromise

Option4 electronic ionisation and the existing electromagnetic, decay and hadronic constructors remain. On GenericIon, alpha, He3, proton, deuteron and triton, continuous `nuclearStopping` and original low-energy multiple/Coulomb scattering are replaced by TestEm7 ZBL screened elastic scattering.

For affordability, **target recoil energy is deposited locally using TestEm7's native local-deposition treatment; explicit target-recoil cascade tracks are disabled**. Projectile energy loss and deflection are still simulated by screened collisions. This is a primary-ion transport approximation, not the earlier fully tracked cascade. Secondary electrons and secondaries from other registered processes are not globally disabled.

TestEm7's default `G4LindhardRobinsonPartition` splits the local recoil deposit between electronic loss and NIEL. The stored native Geant4 energy channels therefore remain separate from CYGNO's custom TRIM response below.

The cross-section/mean-free-path tables use an energy-dependent transfer resolution parameter, with E the total projectile kinetic energy:

```
Tcut(E) = min(100 eV, max(0.1 eV, 0.001 * E))
```

Projectile energy	Transfer parameter
0.25 keV	0.25 eV
1 keV	1 eV
10 keV	10 eV
100 keV	100 eV
1000 keV	100 eV

This controls approximate cross-section table construction and collision sampling, not an exact sharp lower bound on every sampled transfer. Fine sampling at low energy preserves low-energy direction/range while coarser high-energy sampling reduces actual native steps. ZBL potential coefficients and screening lengths are not fitted.

The constructor low-energy limit remains 0.1 eV, its terminal cutoff 1 eV/u, and the existing charged tracking threshold 25 eV. The maximum-step law remains `cap * min(1,sqrt(E/40 keV))`, without a spatial floor. The NR macros use:

Ion	Original cap (um)	Screening cap (um)
H1	52	208
He4	26	208
C12	13	52
O16	6.5	26
F19	3.25	13

Every saved record is still a native Geant4 step. The prototype's Urban MSC activation at 100 MeV total and screened maximum at 100 MeV/u are retained; their differing energy conventions are not a validated high-energy handoff. Qualification is limited to the stated NR range at or below 1 MeV total.

TRIM response: integrate the existing derivative

The original hard-coded TRIM mean-response curves remain. For `R(E) = E * QF_TRIM(E)`, replace the original finite-step approximation `deposit * dR/dE(Epre)` by:

```
custom_hit_response = R(Epre) - R(max(0, Epre - local_raw_deposit))
```

This integrates the same derivative over the finite deposit. Very small deposits use the midpoint derivative to avoid subtraction loss. The existing sub-25 eV extrapolation remains; a terminal deposit gives `R(Epre)`, matching the mean-QF prescription. ER hits retain raw energy as before.

The earlier screened prototype's fitted 28-node event-gain tables and uniform event correction are removed. No new response fit is needed. Custom scalars and hit vectors are summed through the original event action. Raw and native Geant4 ionising/NIEL-derived channels keep their original meanings.

For a contained primary-only recoil with entirely local deposition, increments telescope to `R(E0)`. Close agreement with the frozen TRIM QF curve is therefore **by construction, not an independent microscopic prediction**. Small departures include effects of separately tracked electronic secondaries. Numerical step-size response fluctuations are removed; an intrinsic NR yield-fluctuation or Fano model is not supplied by this prescription.

Fresh matched comparison

The final test uses **10,000 fresh events per ion per model**, 100,000 events total. Source arrays (energies, positions, directions and IDs) match bitwise. The control is built from an immutable archive of the base commit. Both use the original full ROOT schema and a read-only primary tracking observer.

Single H1, He4, C12, O16 or F19 primaries have log-uniform total energy 0.25-1000 keV and isotropic direction at (0,0,245) mm in the existing 900 mbar, 293.15 K He/CF4 gas. All primary tracks stop inside sensitive gas. Geometry, composition, source distributions and production cuts are unchanged; transport resolution changes are listed above.

The scores equally weight ten logarithmic energy bins. Every event is divided by the frozen TRIM expectation at its own energy. These descriptive mean absolute discrepancies are not significance-weighted goodness-of-fit tests. They are not directly interchangeable with the earlier five-energy scores.

Ion	Endpoint error original / screening (%)	Projected error original / screening (%)	Mean cosine error original / screening
H1	34.52 / 5.77	37.96 / 3.30	0.0711 / 0.0183
He4	21.10 / 21.23	23.73 / 20.10	0.0349 / 0.0094
C12	48.52 / 12.52	54.69 / 13.04	0.1684 / 0.0149
O16	51.00 / 11.51	58.81 / 12.56	0.2165 / 0.0156
F19	51.80 / 9.88	60.24 / 10.99	0.2407 / 0.0169

Endpoint agreement improves for H/C/O/F; He endpoint agreement is essentially unchanged. Bending improves in these descriptive scores. Residual errors remain, and average improvement does not imply agreement at every energy. Endpoint bending concerns displacement relative to the initial direction, not final momentum or a reconstructed detector angle.

In the highest energy bin (geometric means about 0.66 MeV), endpoint/TRIM ratios are H 1.009, He 1.164, C 0.695, O 0.696 and F 0.749. C/O endpoints therefore remain approximately 30% short and F about 25% short in that bin.

The original sample has 235/50,000 raw sensitive-gas deposits above `E0 * 1.0001`, reaching 8.12% excess in the largest F event. The new sample has zero such events, with no raw-energy rescaling. Continuous nuclear stopping was implicated in the original budget excess during earlier diagnostics.

File-size qualification

Actual 10,000-event files, scaled to one million events:

Ion	Original GB/million	Screening GB/million
H1	9.638	8.786
He4	9.217	9.494
C12	9.335	7.545
O16	8.955	7.335
F19	8.923	7.354

All new pilot projections are below 10 GB; He has about 5% margin. These are measurements of full-schema files followed by sample-size extrapolation, **not measured million-event files or an arbitrary-spectrum size guarantee**. A high-energy-concentrated sample or changed caps requires another size check. No output compaction or special decoder contributes to these numbers. Mean native hit counts are approximately 78, 82, 65, 63 and 63 for H/He/C/O/F. Step counts are not a wall-time benchmark.

Checks, uncertainty and limits

All ROOT branch names and types match the original. All six event energy channels match their hit-vector sums. IDs, vector lengths, raw budgets and contained stopping are checked for all 50,000 new events. Source arrays match bitwise between models. Both ordinary and observed production targets build.

The frozen TRIM range reference has 195 energy/species points from 897 batches and 89,700 ions. Range ratio errors include finite Geant4 statistics and conditional TRIM mean uncertainty, excluding interpolation/model systematics. TRIM mixture spread curves include the variation of energy inside each bin. Backward fractions use Wilson 95% intervals. Geant4 true curved path is not equated with the sampled TRIM polyline.

Local target-energy deposition omits explicit secondary-recoil branches and their spatial correlations. Good primary ranges/bending and imposed TRIM mean QF do not independently validate charge-cloud morphology, head-tail efficiency, reconstructed angular resolution or intrinsic yield fluctuations. The report compares charge-axis/head-tail quantities between models; the frozen reference does not supply per-event 3D charge clouds for their TRIM validation. A detector response assessment remains necessary.

Code inventory and build

- `src/CYGNOPhysicsList.cc`: screened ion transport replaces overlapping nuclear stopping/scattering; target recoil energy is locally deposited.
- `external/TestEm7/G4ScreenedNuclearRecoil.cc`: adaptive transfer parameter, plus the existing ZBL quadrature continuation and consistent terminal-energy fixes. Other required TestEm7 sources/headers and license are retained.
- `src/CYGNOHit.cc`: finite-step integral of the original TRIM response.
- Five `macro/nuclear_recoils_4pi/*.mac` files: revised maximum-step caps.
- `CMakeLists.txt`: compile two vendored TestEm7 implementation units.
- The former event-calibration header is removed. ROOT writing, hit creation, event output and primary generation have no changes from the original branch.

```
cmake -S . -B build_screening -DWITH_GEANT4_UIVIS=OFF
cmake --build build_screening --parallel 3
```

Use the existing ROOT reader and the branch's NR macros. Switching branches requires a rebuilt or separate executable. Production has no scan environment variables or compact-output mode.

Upstream: Geant4 v11.4.0 TestEm7. Original notices and license are retained in `external/TestEm7`.